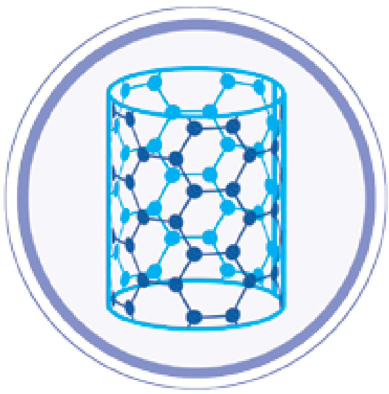
This Philips S9000 electric shaver uses nanotechnology in the blades

better, it's also used as a UV barrier to increase the shelf-life of foods that can spoil with light – such as probiotic yoghurt with live cultures of good bacteria that UV radiation would otherwise kill.

And this UV protection is not just for your probiotic foods, it's also used in sunscreen to help protect your skin from UV radiation. If you've been to the beach and put on sunscreen, you've used nanotech to stay safe. In fact, the "beauty industry" as a whole relies on nanotechnology for a lot of their most popular products. Sun protection aside, there are lipsticks that stay on all day, sealers, concealers, anti-aging skin creams, shampoos and conditioners that promise healthier and shinier hair, etc. Shaving razors also use nanotechnology (as a coating to prevent rusting and keeping the edge sharp) in their manufacturing processes. All this means that pretty much anyone alive today has had some sort of run in with nanotech already, and without even knowing it!

This reliance of nanotechnology to make our lives even better has not plateaued, and will in fact only increase to the point where almost all improvements in our life will be nanotech driven. In fact, soon no one will focus on the nanotechnology aspect anymore, and will just take it for granted as merely another tool used to improve something – much like we take computing for granted today.



NANOMATERIALS

A nanomaterial is really just any material that has at least one dimension that is at the nano scale – between 1 and 100 nm in size, as we mentioned earlier in this article. This means that something that is less than 100 nm thick, or long or wide or tall, or all of the above even, is considered to be a nanomaterial. Of course, if something is nano-sized in all dimensions, it's essentially a nanoparticle, not a material, but more on this later.

Why do we need materials to be nano-sized anyway though? To understand this, you need to first understand what was wrong with materials before nanotechnology came along. Larger dimensions of materials means larger things in general. When it was only possible to use animal hides to make shoes, they were bulkier, harder to work with, and far less comfortable. Imagine the amount of hunting our ancestors from 10,000 years ago (that's the oldest shoe we know of, though humans might have been wearing shoes 40,000 years ago!) could have done with a modern pair of shoes?

Not just shoes, imagine what our ancestors could do with a modern bow and arrow set? Today, thanks to carbon fibre bows and arrows, an archer is able to achieve as much lethal force (if not more), with a kit that is far lighter, and with far less strength required to operate. Let's look at how nanomaterials improve designs in various ways.

Surface area: One of the main ways in which nanomaterials are beneficial is because of the miniaturisation process itself. Surface area plays a huge role in how reactive a material is, and increasing surface area (per given volume of material) is exactly what happens when working with the nanoscale. A simple example is to consider the surface area to volume ratio of particles (let's consider them to be spheres for ease of calculation). We know that the surface area of a sphere is $4\pi r^2$ and volume is $4/3\ \pi r^3$. Divide both sides by $4\pi r^2$, and we get a ratio of 1:3/r for surface area to volume. Thus, as the radius of the particles decreases the ratio increases – for example, for a set volume of material with particles of a radius of 3 nm (0.000000003 m) has a ratio of 3 / 0.000000003 = 1,000,000,000;

Next gen nanomaterials

This is a whole special issue of the journal Molecular Structure, exploring a number of talks in a nanomaterials conference, exploring the next generation of promising materials.
http://dgit.in/nanomats

Hunting viruses

As nanotechnologies are as small as antibodies, they can be used to effectively screen viruses, not only as entire particles, but also only their DNA or particular protein structures.
http://dgit.in/nanovirus

2D materials

Exploring the unusual properties of sheets of material only a single molecule in thickness, and how these nanomaterials can best be used to serve the needs of man.
http://dgit.in/nano2d